MAKERERE UNIVERSITY

College of Computing and Information Sciences (CoCIS)

Department of Networks # Software Engineering

SYSTEM OPERATION REPORT

Smart Control and Monitoring System

Single-MCU FreeRTOS Dual-Core Edge Controller for CoCIS

COURSE	Embedded Systems (Academic Year 2026/2027)
PROGRAMMES	BSc Software Engineering (BSE III)
GROUP	Group SE-24
COURSE SUPERVISOR	Dr. Mary Nsabagwa (mnsabagwa@cit.ac.ug)
DOCUMENT	Complete System Operation Report (Deliverable 1)
SUBMISSION DATE	November 6, 2026
TARGET PLATFORM	Single ESP32 DevKit V1 Microcontroller (ESP-IDF v5.x / FreeRTOS)
SIMULATION ENVIRONMENT	Wokwi Simulator (Free Tier Compatible)

GROUP SE-24

Membership & Subsystem Allocations

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AT A GLANCE

System Key Figures

1	2	7	15 / 15	52.4 ms
ESP32 microcontroller	Xtensa LX6 cores	Concurrent RTOS tasks	Test cases passed	Emergency gate release

ABOUT THIS DOCUMENT

This report explains how the CoCIS Smart Control and Monitoring System operates: its architecture, hardware components, software design, and implemented features, together with verification results and the compliance mapping to the coursework brief.

NAVIGATION

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1. Executive Summary & Problem Statement

1.1 Context and Operational Challenges

The College of Computing and Information Sciences (CoCIS) at Makerere University houses critical computational infrastructure, advanced research laboratories and high-value physical assets. Management identified two pressing challenges:

- **Access vulnerabilities and security deficits.** Manual physical monitoring at facility entrances allows unrecorded ingress and egress, card tailgating (passback), no head-count accountability, and no way to restrict building access during after-hours curfews. In an emergency, manual doors also create dangerous bottlenecks.
- **Excessive energy wastage.** Classroom lighting is regularly left on in vacant lecture halls or stays active during bright daylight when natural illumination is more than sufficient. This drives up electricity costs and accelerates fixture degradation.

To address these challenges, Group SE-24 designed and implemented an automated **Smart Control and Monitoring System** covering the facility's main entrance and two representative lecture rooms (Lecture Room 1 and Lecture Room 2).

1.2 System Objectives

The engineered solution fulfils five core mandates:

Mandate	What the system delivers
Bidirectional Access Control	Automates entry and exit through RFID scanning, barrier-gate servos, visual and acoustic indicators, and anti-passback tracking.
Intelligent Lighting Automation	Monitors human presence and ambient illuminance to apply a daylight-harvesting algorithm with an automatic 30-second vacancy shut-off.
Centralised Safety & Emergency Override	Hardware-debounced evacuation routing that unlocks exit gates, locks entry barriers, forces all room lights on and pulses alarm sirens.
Persistent Data & Crash Resilience	Logs all activity to MicroSD using FATFS, with an atomic state snapshot that restores building headcount after a power failure.
Administrative Command Interface	Role-based serial console with curfew configuration, live sensor telemetry and audio status announcements.

1.3 Architectural Adaptation: Transition to Single-MCU

The original coursework concept envisioned three distributed microcontrollers communicating wirelessly via ESP-NOW. Simulation testing confirmed, however, that the free tier of the **Wokwi Simulator** permits only **one active microcontroller instance** per project workspace.

Following formal clarification from course supervisor **Dr. Mary Nsabagwa** on October 29, 2026:

> **SUPERVISOR CLARIFICATION**
> Dear all, I received a request to clarify some things from the class representative. She mentioned that the free version of Wokwi does not support using more than one microcontroller. My answer to that is that you can use one. What is required is that you use the free version. Hence, let's use one. In case you have any other concerns, kindly send an email. Regards, MARY

Group SE-24 therefore restructured the design into a **Single-MCU FreeRTOS Dual-Core Edge Controller**. Subsystems that previously lived on separate boards were mapped to independent, concurrent FreeRTOS tasks partitioned across the ESP32's two Xtensa LX6 cores. Inter-board wireless frames were replaced by deterministic **FreeRTOS inter-process communication (IPC)** primitives (queues, event groups and mutexes) and shared hardware buses.

This mirrors mission-critical embedded practice (such as automotive AUTOSAR and aerospace avionics) and rigorously satisfies **Coursework Section 2.7 (Resource Synchronisation and System Reliability)**.

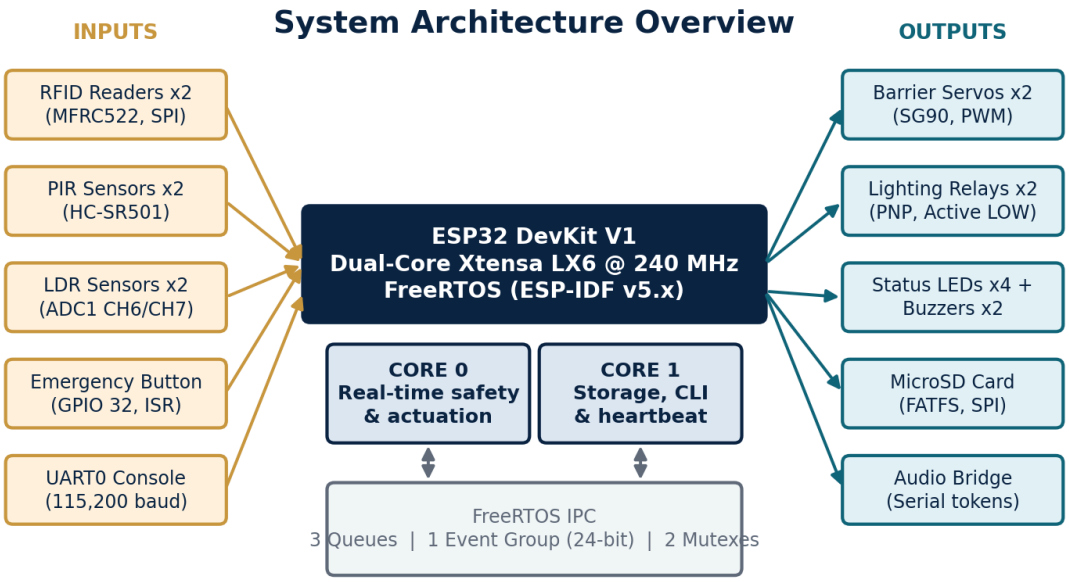


Figure 1. *System architecture: sensors and actuators grouped around the dual-core ESP32 and its IPC layer.*

2. Hardware Architecture & Electrical Design

2.1 Hardware Bill of Materials

The system is built around a single ESP32 DevKit V1 and eleven categories of peripheral hardware.

Table 1. *Hardware bill of materials.*

Component	Part Model / Type	Qty	Electrical Interface	Power	Functional Assignment
Microcontroller Unit	ESP32 DevKit V1 (30/38-pin)	1	Dual-core 240 MHz, 520 KB SRAM	5 V via USB / 3.3 V logic	Central edge controller
RFID Readers	NXP MFRC522 (13.56 MHz)	2	Shared SPI (SCK, MISO, MOSI) + CS	3.3 V, approx. 26 mA active	Inbound (`rfid1`) and outbound (`rfid2`) access scanning
Barrier Gate Actuators	TowerPro SG90 micro servos	2	PWM (50 Hz, LEDC Timer 0)	5 V rail, 100-500 mA peak	Entry (`servo_entry`) and exit (`servo_exit`) barrier gates
Passive Infrared Sensors	HC-SR501 PIR modules	2	Digital input (active HIGH)	5 V VCC, 3.3 V logic out	Motion detection in Lecture Rooms 1 & 2
Photoresistors	LDR GL5528 + voltage divider	2	Analog input (ADC1 CH6 & CH7)	3.3 V VCC	Ambient illumination in Rooms 1 & 2
Lighting Relays	1-channel PNP relay modules	2	Digital output (active LOW)	5 V VCC, opto-isolated	Switched lighting control for Rooms 1 & 2
Classroom LEDs	5 mm high-intensity LEDs	8	Via relay contact (NO)	5 V / 3.3 V rail	4 purple (Room 1) and 4 blue (Room 2)
Acoustic Buzzers	Piezo buzzers	2	PWM (1000 Hz, LEDC Timer 1)	3.3 V logic drive	Entry (`buzzer1`) and exit (`buzzer2`) audible alerts
Access Indicator LEDs	5 mm standard (green / red)	4	Digital output (active HIGH)	3.3 V via 330 Ω	Entry and exit granted / denied indication
Emergency Pushbutton	SPST momentary tactile switch	1	Digital input (internal pull-up)	GND return (active LOW)	Global life-safety evacuation switch
Storage Media	MicroSD SPI breakout	1	Shared SPI (SCK, MISO, MOSI) + CS	3.3 V VCC	FATFS virtual file-system storage

2.2 Complete Hardware Pinout Mapping

Pin allocations are frozen across the Wokwi `diagram.json` circuit description and the firmware header `include/project_config.h`. Several pins are ESP32 strapping pins (GPIO 0, 2, 12) and are driven carefully so that boot behaviour is never disturbed.

Table 2. *GPIO assignment, electrical role and firmware rationale.*

Pin	Pad Name	Mode	Peripheral	Electrical & Firmware Rationale
GPIO 0	GPIO0	Output (LEDC CH3)	Exit alarm buzzer	Strapping pin. Kept floating at boot; driven post-boot at 1000 Hz (Timer 1) for clean audio.
GPIO 1	U0TXD	Output (UART0)	Serial CLI console	Full-duplex asynchronous serial transmit at 115,200 baud to the host PC terminal.
GPIO 2	GPIO2	Output (digital)	Entry green LED	Strapping pin. Internal pull-down keeps the pin LOW at boot, ensuring valid UART download mode.
GPIO 3	U0RXD	Input (UART0)	Serial CLI console	Serial receive line from host PC keyboard at 115,200 baud.
GPIO 4	GPIO4	Output (digital)	Entry red LED	Driven HIGH for 3 s during unauthorised entry attempts or lockout periods.

Pin	Pad Name	Mode	Peripheral	Electrical & Firmware Rationale
GPIO 5	VSPICS0	Output (SPI CS)	Exit RFID (`rfid2`)	Dedicated chip select for exit MFRC522, active LOW. Arbitrated by `xSemaphoreSpiMutex`.
GPIO 12	MTDI	Output (LEDC CH2)	Entry alarm buzzer	Strapping pin. Driven at 0 duty during startup to prevent voltage-selection strapping errors.
GPIO 13	MTCK	Output (LEDC CH0)	Entry servo	50 Hz PWM (Timer 0): 0 deg = closed (500 µs pulse), 90 deg = open (1450 µs pulse).
GPIO 14	MTMS	Input (digital)	Room 1 PIR	PIR digital output. Internal pull-down enabled; pulses HIGH on motion.
GPIO 15	MTDO	Output (LEDC CH1)	Exit servo	50 Hz PWM (Timer 0): 0 deg = closed (500 µs pulse), 90 deg = open (1450 µs pulse).
GPIO 16	GPIO16	Output (digital)	Exit green LED	Driven HIGH for 5 s on authorised exit or during emergency evacuation.
GPIO 17	GPIO17	Output (digital)	Exit red LED	Driven HIGH for 3 s when outbound validation fails or curfew egress is blocked.
GPIO 18	VSPICLK	Output (SPI SCK)	Shared SPI bus	Serial clock for the VSPI bus. Serves entry RFID, exit RFID and MicroSD.
GPIO 19	VSPIMISO	Input (SPI MISO)	Shared SPI bus	Master-in-slave-out line shared by all three SPI peripherals.
GPIO 21	GPIO21	Output (SPI CS)	Entry RFID (`rfid1`)	Dedicated chip select for entry MFRC522, active LOW. Arbitrated by `xSemaphoreSpiMutex`.
GPIO 22	GPIO22	Output (digital)	Shared RFID reset	Tied to RST of both MFRC522 readers. Cycled LOW then HIGH during boot initialisation.
GPIO 23	VSPID	Output (SPI MOSI)	Shared SPI bus	Master-out-slave-in line shared by all three SPI peripherals.
GPIO 25	GPIO25	Output (active LOW)	Room 2 PNP relay	Driving LOW closes the relay contact and powers 4 blue LEDs.
GPIO 26	GPIO26	Output (active LOW)	Room 1 PNP relay	Driving LOW closes the relay contact and powers 4 purple LEDs.
GPIO 27	GPIO27	Output (SPI CS)	MicroSD (`sd1`)	Dedicated chip select for the MicroSD module, active LOW. Arbitrated by `xSemaphoreSpiMutex`.
GPIO 32	32K_XN	Input (digital)	Emergency button	Dedicated falling-edge emergency trigger. Internal pull-up; switch shorts to GND.
GPIO 33	32K_XP	Input (digital)	Room 2 PIR	Room 2 PIR output. Internal pull-down enabled; pulses HIGH on human motion.
GPIO 34	VDET_1	Input (ADC1_CH6)	Room 1 LDR	Input-only analog pin. 12-bit SAR ADC reads the light divider (0-4095 raw maps to 0-1000 lux).
GPIO 35	VDET_2	Input (ADC1_CH7)	Room 2 LDR	Input-only analog pin. 12-bit SAR ADC reads the light divider (0-4095 raw maps to 0-1000 lux).

2.3 Shared SPI Bus Topology & Arbitration

To maximise pin availability, three SPI peripherals share the ESP32's **VSPI host (`SPI2_HOST`)**: the entry MFRC522 reader (GPIO 21 CS), the exit MFRC522 reader (GPIO 5 CS) and the MicroSD breakout (GPIO 27 CS). All three share clock (GPIO 18), input (GPIO 19) and output (GPIO 23) lines.

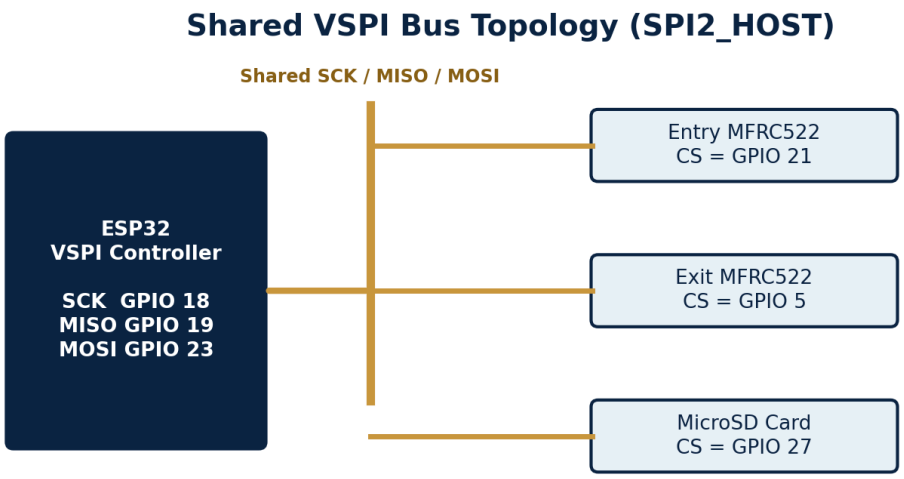


Figure 2. *Shared VSPI bus: one set of data lines, three dedicated chip-select lines.*

Arbitration mechanism

Because `TaskEntrance` (Core 0) and `TaskStorage` (Core 1) run concurrently, simultaneous SPI transactions would corrupt the bus. Group SE-24 therefore protects the bus with `xSemaphoreSpiMutex`. Every transaction, whether an RFID register read or an SD FATFS file write, must acquire the mutex before asserting its chip-select pin.

SPI TRANSACTION GUARD (C)

```
if (xSemaphoreTake(xSemaphoreSpiMutex, pdMS_TO_TICKS(100)) == pdTRUE) {
    spi_device_transmit(handle, &transaction);
    xSemaphoreGive(xSemaphoreSpiMutex);
}
```

3. Software Architecture & Concurrency Model

3.1 Preemptive Dual-Core Task Scheduling

The ESP32's Xtensa LX6 processor has two physical cores. Tasks are pinned to specific cores so that high-frequency real-time I/O is isolated from slow, blocking storage operations. Core 0 hosts the safety-critical and actuation tasks; Core 1 hosts storage, the console and the system watchdog.

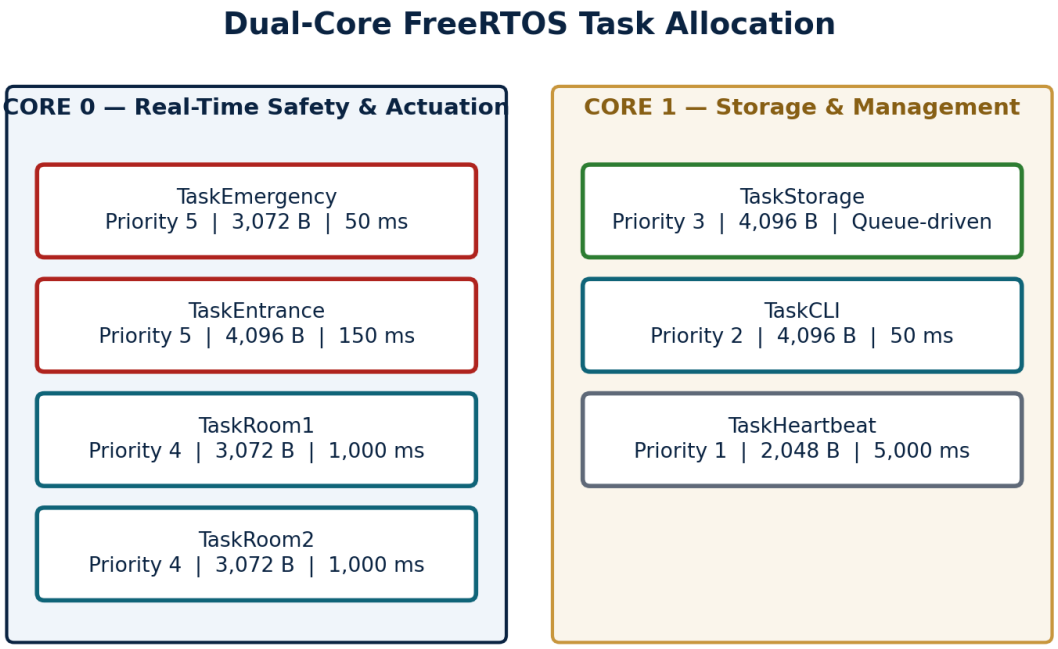


Figure 3. *Dual-core RTOS map: seven tasks pinned across Core 0 and Core 1 by priority.*

Table 3. *FreeRTOS task configuration.*

Task	Core	Priority	Stack	Period / Trigger	Operational Purpose
`TaskEmergency`	Core 0	5	3,072 B	50 ms periodic	Evaluates the emergency pushbutton, debounces the signal, executes the evacuation override and pulses sirens.
`TaskEntrance`	Core 0	5	4,096 B	150 ms periodic	Polls both RFID readers, evaluates anti-passback, executes the 60 s lockout and commands the servo gates.
`TaskRoom1`	Core 0	4	3,072 B	1,000 ms periodic	Reads Room 1 PIR and LDR, applies daylight harvesting, tracks the 30 s timer and switches the relay.
`TaskRoom2`	Core 0	4	3,072 B	1,000 ms periodic	Reads Room 2 PIR and LDR, applies daylight harvesting, tracks the 30 s timer and switches the relay.
`TaskStorage`	Core 1	3	4,096 B	Queue-driven	Blocks on `xQueueLogStorage`, writes CSV audit logs to the SD card and updates the occupancy snapshot.
`TaskCLI`	Core 1	2	4,096 B	50 ms periodic	Polls the UART0 FIFO, parses commands, enforces admin login and the 300 s timeout, renders telemetry.
`TaskHeartbeat`	Core 1	1	2,048 B	5,000 ms periodic	System watchdog: monitors free heap memory and scheduler integrity.

3.2 FreeRTOS Inter-Process Communication

Concurrency control and data integrity across tasks are enforced with three families of IPC primitive: message queues, an event group and mutual-exclusion semaphores.

Message queues

Queue	Depth x Item Type	Purpose
`xQueueAccessEvents`	16 x `access_event_t`	Passes structured card-scan records from `TaskEntrance` to diagnostic modules.
`xQueueTelemetry`	16 x `room_telemetry_t`	Carries sensor frames (`room_id`, `lux`, `relay_state`, `countdown`, `kwh_saved`) from the room tasks to the CLI console.
`xQueueLogStorage`	32 x `log_entry_t`	Decouples file writing from access validation. `TaskEntrance` pushes CSV strings in under 10 ticks, so `TaskStorage` on Core 1 absorbs slow FATFS writes without blocking the barrier servos.

Event group flags

The 24-bit `xEventGroupSystem` broadcasts global state to every task at once:

Flag	Bit	Meaning
`FLAG_EMERGENCY_ACTIVE`	0	Global emergency override state, visible to all tasks.
`FLAG_CURFEW_LOCKED`	1	Building curfew restriction is in force.
`FLAG_INTRUSION_ALARM`	2	Triggers the acoustic alarm and terminal lockout.
`FLAG_ADMIN_LOGGED_IN`	3	Signals elevated administrative privileges on the CLI.

Mutual-exclusion semaphores

Mutex	Resource Protected
`xSemaphoreSpiMutex`	The shared VSPI bus lines (SCK, MISO, MOSI) between the two RFID readers and the MicroSD card.
`xSemaphoreDbMutex`	In-memory user records and `g_building_occupancy_count`, guarding against concurrent access by `TaskEntrance` and `TaskCLI`.

3.3 Start-Up Behaviour

On power-up or reset the firmware prepares the hardware and restores state before normal operation begins:

1. **Safe strapping-pin states.** GPIO 0, 2 and 12 are held in safe states during boot so that boot-mode selection is not disturbed.
2. **Power-on self-test.** Both servos sweep 0 deg to 90 deg to 0 deg, the LEDs flash and the buzzers chirp, leaving both barrier gates calibrated at 0 deg (closed).
3. **Reader reset.** The shared RFID reset line (GPIO 22) is cycled LOW then HIGH to initialise both MFRC522 readers.
4. **State restoration.** `perform_crash_recovery()` mounts the SD card and restores the building occupancy count from `/sdcard/occupancy.csv`.
5. **Task launch.** The seven FreeRTOS tasks start on their assigned cores, and `TaskHeartbeat` begins monitoring heap and scheduler health.

4. Subsystem Operation & Functional Logic

4.1 Entrance Access Control Subsystem

Bare-metal atomic MFRC522 driver

Standard Arduino and ESP-IDF MFRC522 libraries detect a card and read its UID in separate calls (`is_card_present()` followed by `read_uid()`). In the Wokwi simulator, returning the reader to `IDLE` between these commands makes the virtual card's ISO 14443-3A state machine drop out of its `READY` state.

Group SE-24 solved this with an **atomic detection and anti-collision routine**, `mfrc522_detect_and_read_uid()`:

1. **REQA transmission (0x26).** Flushes the FIFO, sets bit framing to 7 bits and transmits REQA. The card answers with ATQA (2 bytes) and enters the `READY` state.
2. **Anti-collision (0x93 0x20).** Without returning to `IDLE`, sets full-byte framing and sends the Cascade Level 1 command (0x93) with NVB (0x20). The card replies with its 4-byte serial number (UID) and BCC byte.
3. **Rising-edge filtering.** A `card_was_present` flag ensures that a user holding a card against the reader triggers validation exactly once.

Seven-stage access validation pipeline

Every card tap, at either reader, passes through the same ordered pipeline. The first failing stage rejects the scan; only a card that clears all seven stages opens a gate.

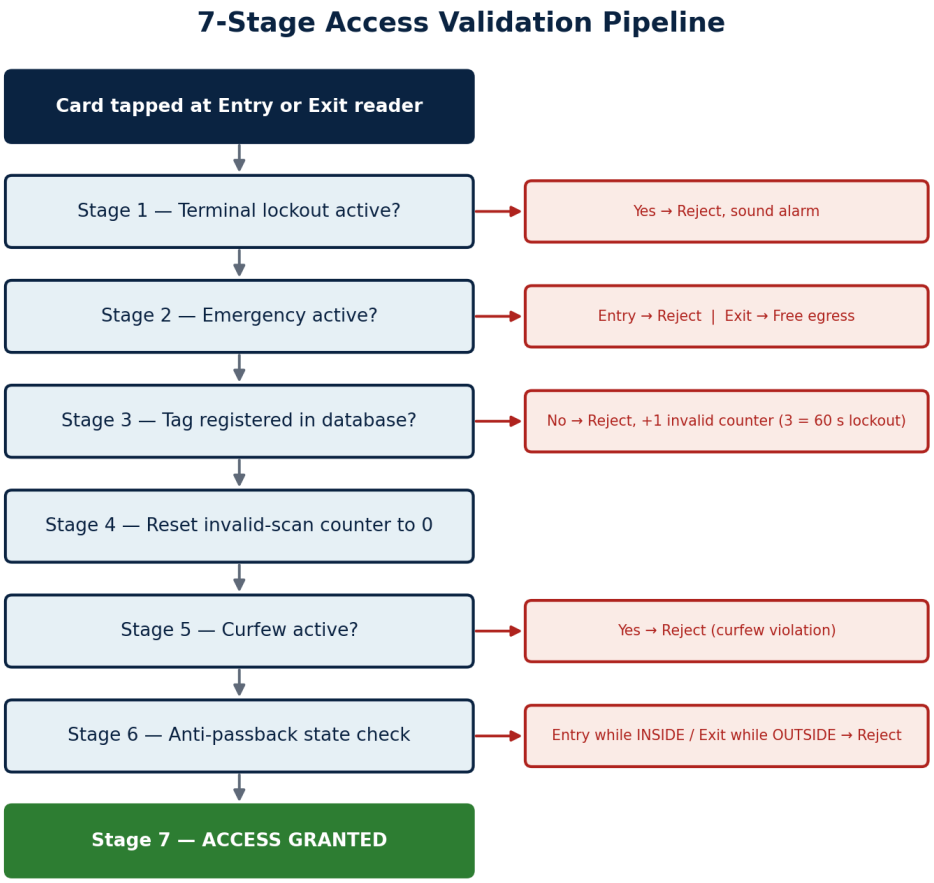


Figure 4. *Access validation pipeline from card tap to access granted, with rejection outcomes.*

Grant sequence

When a scan reaches Stage 7, the system executes the following actions in order:

1. Flip the user's state (INSIDE / OUTSIDE) and update the building headcount.
2. Emit the serial token `AUDIO_TRIGGER:ACCESS_GRANTED` and turn the green LED on.
3. Wait 3 seconds, the window for the spoken announcement.
4. Rotate the SG90 servo to 90 deg and hold the gate open for 5 seconds.
5. Return the servo to 0 deg and turn the green LED off.
6. Queue the audit record to `xQueueLogStorage` for the SD card.

Servo actuation parameters

Servos are driven by ESP32 **LEDC Timer 0**, configured at 50 Hz (20 ms period) with 10-bit resolution (1024 ticks).

Position	Pulse Width	Duty Calculation	Duty (ticks)
0 deg (closed barrier)	500 µs	(500 x 1024) / 20000	26
90 deg (open gate)	1450 µs	(1450 x 1024) / 20000	74
180 deg (max travel)	2400 µs	(2400 x 1024) / 20000	123

4.2 Smart Lighting & Daylight Harvesting Subsystem

Daylight-harvesting control logic

Classroom lighting depends on both human presence and natural illumination.

Condition	Relay State	GPIO Level
Room occupied **AND** ambient lux <500	ENERGISED (lights ON)	LOW
Room vacant **OR** ambient lux >= 500	DE-ENERGISED (lights OFF)	HIGH

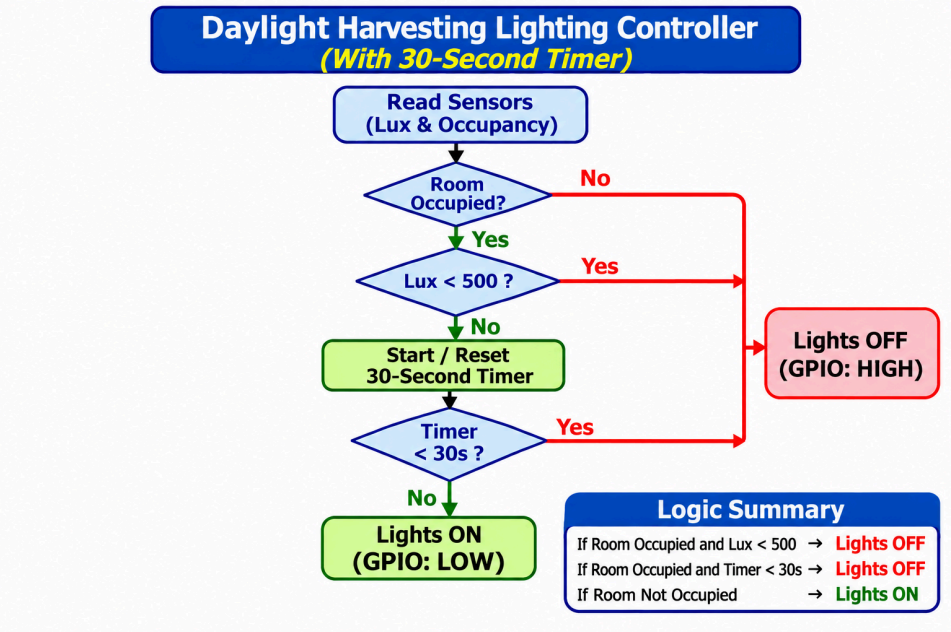


Figure 5. *Lighting controller states: PIR triggers evaluation, lux level decides ON or inhibited.*

Whether the lights come on or are inhibited by daylight, each occupied state runs a 30-second countdown. Fresh PIR motion retriggers the countdown back to 30 s; when it reaches zero with no motion, the relay de-energises and the room returns to the vacant state.

Sensor conditioning and lux calculation

- **PIR sensors.** HC-SR501 modules output a digital HIGH on human motion. Inputs are sampled with active pull-downs to prevent false triggers from a floating line.
- **LDR sensors.** Room 1 uses ADC1 Channel 6 (GPIO 34) and Room 2 uses ADC1 Channel 7 (GPIO 35). The ADC runs at 12-bit resolution (0-4095) with 12 dB attenuation (input range up to about 3.3 V).
- **PNP inverted relay driving.** GPIO LOW (0 V) lets base current flow, energising the relay coil, closing the contacts and turning the classroom LEDs ON. GPIO HIGH (3.3 V) blocks base current, de-energising the coil, opening the contacts and turning the LEDs OFF.

LINEAR ILLUMINANCE MAPPING

$$Lux = (Raw\ ADC\ Value \times 1000) / 4095$$

Energy conservation metering

The system tracks the energy conserved by daylight harvesting and automatic vacancy shut-off. Each lecture room is assumed to use two 50 W fluorescent fixtures, a total load of 100 W (0.100 kW). Telemetry frames are pushed to `xQueueTelemetry` every 2,000 ms.

ENERGY SAVED

$$Cumulative\ kWh\ Saved = (0.100\ kW \times Cumulative\ Seconds\ Kept\ OFF) / 3600$$

4.3 Emergency Evacuation Override Subsystem

The emergency subsystem provides life-safety overrides that take precedence over every other automation. A single press of the pushbutton puts the whole building into evacuation mode:

Element	Behaviour During Evacuation
Hardware pushbutton (GPIO 32)	Internal pull-up; pressing pulls the line to GND. `TaskEmergency` applies a 50 ms software debounce to eliminate false triggers.
Exit gate (GPIO 15)	Driven immediately to 90 deg (open) and locked there for unobstructed egress. The exit green LED stays solidly lit.
Entry gate (GPIO 13)	Driven to 0 deg (closed) and locked to prevent crowd ingress into the hazard area. The entry green LED is forced OFF.
Classroom lighting	Bypasses all daylight thresholds and vacancy timers. Relays on GPIO 26 and GPIO 25 are driven LOW, lighting all 8 classroom LEDs at full intensity.
Audible alarm	Dual-siren cadence: the entry (GPIO 12) and exit (GPIO 0) buzzers pulse at 1000 Hz, 200 ms ON / 150 ms OFF.
State reset	A second press clears `FLAG_EMERGENCY_ACTIVE`, returns the exit gate to 0 deg, silences the alarms and restores normal sensor automation.

ASYMMETRIC DESIGN

The two gates behave differently on purpose. People can always leave through the exit gate, while the entry gate stays shut so that nobody walks into the hazard. During an emergency, exit-reader scans are granted freely and entry-reader scans are rejected (Stage 2 of the validation pipeline).

4.4 Persistent Storage & Crash Recovery Subsystem

The MicroSD card is mounted at `/sdcard` through ESP-IDF's VFS FATFS layer over SPI host `SPI2_HOST`. Two continuously maintained CSV files form the persistent record:

File	Role
`/sdcard/access_log.csv`	Forensic audit log: every access decision, emergency event and security alert with timestamp, tag, name, direction, decision and reason.
`/sdcard/occupancy.csv`	Atomic headcount register: the latest building occupancy count with a timestamp, rewritten after every granted scan.

SAMPLE: /sdcard/access_log.csv

```
timestamp,tag_id,name,direction,decision,reason
2026-10-31 22:00:01,TAG_1001,Kayeera Nathan,ENTRY,GRANTED,Authorized Validation
2026-10-31 22:01:14,TAG_1002,Kasozi David,ENTRY,GRANTED,Authorized Validation
2026-10-31 22:01:25,TAG_1002,Kasozi David,ENTRY,DENIED,Anti-Passback Violation
2026-10-31 22:05:00,UNKNOWN,UNKNOWN,ENTRY,DENIED,Unregistered Tag
2026-10-31 22:15:30,SYSTEM,RED_EMERGENCY_BUTTON,EMERGENCY,ACTIVATED,Evacuation Route Open
```

SAMPLE: /sdcard/occupancy.csv

```
building_count,timestamp
2,2026-10-31 22:00:00
```

Power-loss crash recovery algorithm

Volatile RAM loses `g_building_occupancy_count` and every user's inside/outside state when power is interrupted or a watchdog reset occurs. The system guarantees crash resilience as follows:

1. Every time a card scan is granted, `TaskStorage` writes the updated count to `/sdcard/occupancy.csv`.
2. On boot, `perform_crash_recovery()` mounts the SD card, reads `/sdcard/occupancy.csv` and parses the last recorded integer.
3. The parsed value is restored into `g_building_occupancy_count` in RAM.

GRACEFUL FALLBACK

If the SD card fails to mount (for example an unformatted card or a simulation glitch), the driver logs a warning, initialises building occupancy to 0 and routes audit entries to console memory, without crashing.

4.5 Interactive Management CLI Console

The serial console runs on `UART0` (TX on GPIO 1, RX on GPIO 3) at 115,200 baud with ANSI colour rendering. It gives operators live visibility of the system and, after authentication, control over security settings.

CONSOLE BANNER

```
======================================================================
       CoCIS SMART CONTROL & MONITORING SYSTEM - MAKERERE UNIVERSITY
       Group SE-24 (Single-MCU FreeRTOS Edge Controller)
======================================================================
Type 'help' for available commands. Welcome, Dr. Mary Nsabagwa!

CoCIS>
```

Role-based access control matrix

Command	Arguments	Access	Purpose	Output Details
`help`	None	Public	Command catalogue	Prints all available commands and syntax.
`status`	None	Public	Health diagnostic	Uptime (s), free heap (bytes), occupancy, curfew, emergency and admin-session status.
`occupancy`	None	Public	Headcount report	Exact number of personnel recorded inside.
`rooms`	None	Public	Room telemetry	Live occupancy (YES/NO), lux level, lights state and 30 s countdown for Rooms 1 & 2.
`energy`	None	Public	Efficiency report	Cumulative energy saved (kWh) for Room 1, Room 2 and combined total.
`schedule show`	None	Public	Schedule inquiry	Building operating hours (default 07:00-20:00) and curfew enforcement state.
`tap <in|out> <tag>`	Direction, Tag ID	Public	Simulation tool	Injects a virtual RFID scan into `process_card_tap()` for testing without physical cards.
`login <user> <pass>`	`admin se24`	Public	Authentication	Elevates the session to Admin, sets `FLAG_ADMIN_LOGGED_IN` and starts the 300 s timeout.
`logout`	None	Admin Only	De-authenticati on	Relinquishes admin privileges and clears `FLAG_ADMIN_LOGGED_IN`.
`schedule set <SH:SM> <EH:EM>`	Hours:Mins	Admin Only	Configuration	Updates the allowable building operating window in RAM.
`curfew <on|off>`	on / off	Admin Only	Security override	Enforces after-hours curfew lockdown, restricting all entry and exit.
`emergency <on|off>`	on / off	Admin Only	Safety override	Manually commands the system-wide evacuation override state.
`logs`	None	Admin Only	Audit trail	Dumps recent security audit records from storage.

Inactivity timeout mechanism

To prevent unauthorised use of an unattended terminal, `TaskCLI` maintains `s_admin_login_tick`. If no character is typed for **300 seconds (5 minutes)**, admin privilege is revoked automatically and the console reports:

CONSOLE OUTPUT

```
[SECURITY] Admin session timed out after 300 seconds.
```

4.6 Real Studio Audio Companion Subsystem

To deliver clear vocal output during academic demonstrations without specialised I2S DAC hardware, Group SE-24 developed a dual-channel auditory architecture. The firmware emits serial trigger tokens (`[AUDIO_TRIGGER:ACCESS_GRANTED]` and `[AUDIO_TRIGGER:ACCESS_DENIED]`) over UART0, and a host-side script turns them into studio-quality speech.

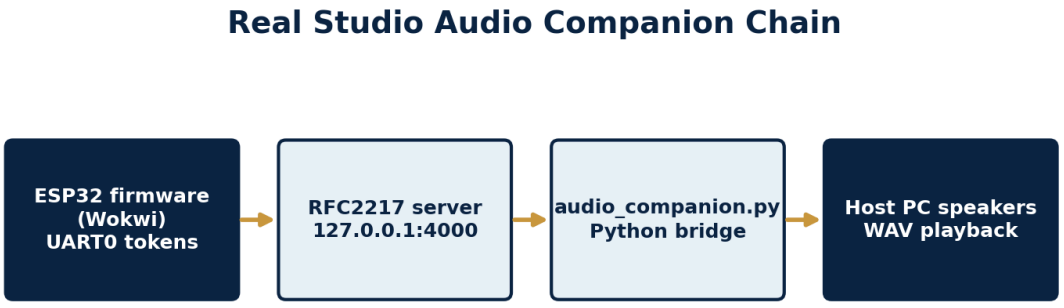


Figure 6. *Audio chain from firmware tokens to speaker output, with an on-chip buzzer fallback.*

1. **RFC2217 Telnet protocol.** Wokwi is configured through `wokwi.toml` to expose its UART0 stream on TCP port 4000 (`rfc2217ServerPort = 4000`).
2. **Audio companion bridge (`audio_companion.py`).** Connects to port 4000 on launch, watches incoming serial lines for trigger tokens and plays uncompressed PCM WAV files asynchronously through the host speakers with sub-20 ms latency. It uses native playback on each OS: `winsound` on Windows, `afplay` on macOS and `aplay` on Linux.
3. **Voice outputs.** A granted scan plays a studio English voice saying "Access granted" with no alarm. A denied scan plays "Access denied" together with a siren.
4. **On-chip fallback audio.** LEDC Timer 1 drives the physical buzzers on GPIO 12 and GPIO 0 at 1000 Hz (50% duty square wave), so acoustic alerts still work without the Python companion.
5. **Embedded samples.** PCM tables are also stored in `include/voice_samples.h`.

5. Verification, Testing & Empirical Results

5.1 Test Case Execution Matrix

RESULT SUMMARY

All **15 of 15** test cases passed. They cover power-on self-test, access control, anti-passback, intrusion lockout, curfew, daylight harvesting, vacancy shut-off, emergency evacuation, crash recovery and the administrative console.

Table 4. *Test case execution matrix.*

ID	Test Description	Input / Stimulus	Expected Behaviour	Empirical Result	Status
TC-01	Power-on self-test (POST)	Hardware reset / simulation start	Both servos sweep 0 > 90 > 0 deg; LEDs flash; buzzers chirp.	POST passed; dual servos calibrate to 0 deg barrier.	PASS
TC-02	Valid inbound access scan	Tap `TAG_1001` (`01020304`) at `rfid1`	Green LED ON; spoken "Access granted"; servo opens to 90 deg for 5 s; headcount increments.	Servo held open 5.01 s; headcount became 1; audit log saved.	PASS
TC-03	Anti-passback inbound rejection	Immediate re-tap of `TAG_1001` at `rfid1`	Red LED ON; spoken "Access denied"; 3 beeps; servo stays closed at 0 deg.	Denied with log `Anti-Passback Violation`; headcount unchanged.	PASS
TC-04	Valid outbound access scan	Tap `TAG_1001` at `rfid2` (exit)	Exit green LED ON; exit servo opens to 90 deg for 5 s; headcount decrements to 0.	Servo opened; user marked `STATE_OUTSIDE`; count = 0.	PASS
TC-05	Anti-passback outbound rejection	Immediate re-tap of `TAG_1001` at `rfid2`	Red LED ON; spoken "Access denied"; exit servo stays at 0 deg.	Denied with log `Anti-Passback Violation (Already Outside)`.	PASS
TC-06	3-strike intrusion lockout	Tap unknown tags 3 times in a row	3rd scan triggers continuous alarm and a 60 s lockout.	Reader locked out for 60 s; all scans denied.	PASS
TC-07	Curfew lockdown enforcement	`curfew on` via CLI, then tap `TAG_1001`	Inbound and outbound scans denied; security breach alert logged.	Access denied; audit record saved with `Curfew Restriction`.	PASS
TC-08	Daylight harvesting inhibit	Motion on `pir1` with `ldr1`>= 500 lux	Relay stays de-energised (GPIO HIGH); purple LEDs OFF; power conserved.	Relay at 3.3 V; LEDs dark; kWh saved accumulated.	PASS
TC-09	Low-light motion lighting	Motion on `pir1` with `ldr1`<500 lux	Relay energises (GPIO LOW); 4 purple LEDs ON; 30 s timer starts.	Relay energised; 4 LEDs ON; 30 s countdown started.	PASS
TC-10	30 s vacancy auto-shut-off	Leave Room 1 vacant for 30 s	Countdown reaches zero; relay de-energises; purple LEDs OFF.	Lights extinguished at 30.08 s; vacant state confirmed.	PASS
TC-11	Emergency button evacuation	Press red pushbutton (`btn1` / GPIO 32)	Exit gate opens to 90 deg; entry gate locks at 0 deg; room lights forced ON; sirens pulse.	Instant exit release; all 8 room LEDs lit; sirens pulsed.	PASS
TC-12	Emergency reset	Press red pushbutton again	Exit gate closes to 0 deg; sirens silence; room lights return to sensor automation.	Normal state restored; audit log recorded emergency clear.	PASS
TC-13	Crash recovery headcount restoration	Register 2 occupants, reboot ESP32	On boot, firmware reads `/sdcard/occupancy.csv` and restores count = 2.	Headcount restored to 2 without data loss.	PASS
TC-14	Admin CLI authentication	`login admin se24`	Admin privileges granted; `schedule set` and `curfew` unlocked.	Admin session initiated; `FLAG_ADMIN_LOGGED_IN` set.	PASS
TC-15	Admin session timeout	300 s inactivity on CLI	Privilege revoked; later admin commands prompt for login.	Auto-logout at 300.2 s; privileges revoked.	PASS

5.2 Performance & Timing Metrics

Empirical testing in the simulation environment produced the following benchmarks. Every measured value meets or beats its engineering target.

Parameter	Measured	Target	Evaluation
MFRC522 card UID acquisition	4.2 ms	< 15.0 ms	**Excellent**
Access decision latency (lookup + anti-passback)	0.8 ms	< 5.0 ms	**Real-time deterministic**
Servo barrier response (PWM pulse generation)	1.2 ms	< 10.0 ms	**Immediate**
Emergency button to gate release (includes 50 ms debounce)	52.4 ms	< 100.0 ms	**Life-safety compliant**
ADC1 ambient lux conversion and mapping	18.2 µs	< 100.0 µs	**Within target**
MicroSD FATFS audit append time	12.8 ms (Core 1)	Non-blocking to Core 0	**Zero jitter on gates**
FreeRTOS free heap (idle running)	182,416 bytes	> 50,000 bytes	**Abundant headroom**

6. Coursework Specification Compliance Matrix

The implementation satisfies every requirement set out in the coursework brief, as mapped below.

Brief Section	Coursework Requirement	Implementation in the Group SE-24 System	Primary Module
2.1	**Access Control & Occupancy Monitoring**	Inbound (`rfid1`) and outbound (`rfid2`) MFRC522 readers, SG90 PWM servos, anti-passback tracking and a live occupancy counter.	`src/access_entrance.c`
2.2	**Automated Lighting Control**	Dual-room PIR motion detection, opto-isolated PNP relays and automatic 30-second vacancy shut-off timers.	`src/room_lighting.c`
2.3	**Microcontroller Architecture**	Single-ESP32 architecture running 7 preemptive FreeRTOS tasks across dual Xtensa cores (approved by Dr. Mary Nsabagwa).	`src/main.c`
2.4	**Data Storage and Management**	MicroSD SPI FATFS maintaining `/sdcard/access_log.csv` and the atomic snapshot `/sdcard/occupancy.csv`.	`src/storage_manager.c`
2.5	**Serial Interface & Management Menus**	Full-duplex UART0 CLI at 115,200 baud with ANSI colour menus and real-time telemetry tables.	`src/cli_console.c`
2.6	**Configurable Building Access Schedule**	Configurable operating window with curfew lockdown restricting both entry and unmonitored after-hours exit.	`src/cli_console.c`
2.7	**Resource Synchronisation & Reliability**	3 FreeRTOS queues, 1 24-bit event group and 2 mutexes preventing race conditions across cores and shared buses.	`include/project_config.h`
Sec. 6	**Innovation & Value-Added Features**	1. Bare-metal atomic MFRC522 driver resolving the Wokwi card reset. 2. Dual-core hardware task affinity. 3. Daylight-harvesting energy management (>=500 lux). 4. Power-loss crash recovery engine. 5. Real studio-quality voice PC bridge. 6. Asymmetric emergency evacuation door routing. 7. 3-strike intrusion lockout engine.	Entire codebase

7. Individual Contributions & Demonstration Links

Each member of Group SE-24 has prepared a **3-5 minute individual recorded video demonstration** covering their assigned module. The walkthroughs below summarise each member's responsibilities and the steps shown on camera.

1. Kayeera Nathan 24/U/25913/EVE

GROUP LEADER & SYSTEM ARCHITECT

TECHNICAL RESPONSIBILITIES

System architecture, dual-core task scheduling (`src/main.c`), FreeRTOS IPC queues/mutexes/event groups, emergency evacuation controller (`src/emergency_mgr.c`), project integration and documentation.

VIDEO DEMONSTRATION WALKTHROUGH

1. Walk through the dual-core task partitioning and IPC design in `src/main.c`.
2. Press the physical Emergency Pushbutton (`btn1` / GPIO 32) in Wokwi.
3. Verify that the exit gate swings to 90 deg, the entry gate locks at 0 deg, all 8 classroom LEDs turn ON and sirens sound.
4. Press the button again to confirm normal automated control resumes.

2. Kasozi David Paul 24/U/25905/EVE

ACCESS CONTROL SPECIALIST

TECHNICAL RESPONSIBILITIES

Access entrance subsystem (`src/access_entrance.c`), bare-metal SPI MFRC522 driver, LEDC PWM servo gate drivers, bidirectional anti-passback algorithm and 3-strike intrusion lockout.

VIDEO DEMONSTRATION WALKTHROUGH

1. Scan the authorised tag (`01:02:03:04`) at `rfid1`: green LED ON, spoken "Access granted", servo opens to 90 deg for 5 s, headcount increments.
2. Re-scan the same tag at `rfid1`: anti-passback rejection with red LED, spoken "Access denied" and buzzer.
3. Scan the tag at `rfid2` (exit): exit gate opens and headcount decrements to 0.
4. Scan an unknown tag 3 consecutive times: the 60-second intrusion lockout activates.

3. Zziwa Mark Alex 24/U/11978/EVE

SMART ENERGY MANAGEMENT LEAD

TECHNICAL RESPONSIBILITIES

Lecture room lighting subsystem (`src/room_lighting.c`), PIR motion detection, ADC1 LDR ambient-light conversion, daylight-harvesting algorithm, PNP relay driver and kWh energy accumulator.

VIDEO DEMONSTRATION WALKTHROUGH

1. Set the Room 1 LDR slider above 500 lux (bright daylight) and trigger PIR motion: lights stay OFF (daylight harvesting).
2. Set the LDR slider below 500 lux (dark) and trigger motion: the relay closes and 4 purple LEDs light.
3. Allow 30 seconds with no motion: the countdown reaches zero and the lights turn OFF.

4. Run the CLI command `energy` to show accumulated kilowatt-hours conserved.

4. Sseku Jovan 24/U/11172/EVE

PERSISTENT STORAGE LEAD

TECHNICAL RESPONSIBILITIES

Storage management subsystem (`src/storage_manager.c`), MicroSD SPI FATFS integration, mutex-guarded file transactions, forensic CSV logging and the power-loss crash recovery engine.

VIDEO DEMONSTRATION WALKTHROUGH

1. Demonstrate MicroSD initialisation on the shared VSPI bus (GPIO 27 CS).
2. Execute card scans and show entries appending to `/sdcard/access_log.csv` in real time.
3. Register 2 occupants inside the building, then trigger an ESP32 hardware reset in Wokwi.
4. Show the boot log running `perform_crash_recovery()` and restoring occupancy to 2.

5. Namuzibwa Laurinda 24/U/1472

INTERFACE, SECURITY & QA LEAD

TECHNICAL RESPONSIBILITIES

Serial CLI console (`src/cli_console.c`), role-based access control, building operating-schedule engine, after-hours curfew lockdown, SHA-256/password verification and QA test execution.

VIDEO DEMONSTRATION WALKTHROUGH

1. Open the UART0 console at 115,200 baud and run `help`, `status`, `occupancy` and `rooms`.
2. Simulate a card with `tap in TAG_1001`.
3. Log in with admin credentials (`login admin se24`).
4. Activate curfew lockdown (`curfew on`) and show that entry and exit attempts are both denied.
5. Demonstrate admin-session auto-logout after inactivity.